

Isonomia: AI-Native Infrastructure for Citable Reasoning

Executive Summary

Language models are good at producing prose about a debate and bad at preserving its structure. They can summarize an argument but lose track of which premise carries the weight, which objection is decisive, which source backs which claim, and which conclusion the evidence does not actually support. They cite web pages, which are the wrong unit: a page mixes claims, evidence, rhetoric, and unstated assumptions into a single document-shaped blob.

Isonomia replaces the page with the argument. Claims, premises, objections, evidence, and the state of a dispute become machine-citable objects — content-hashed, provenance-bearing, standards-compatible, and exposed through web, data, API, and Model Context Protocol surfaces. A system querying Isonomia learns not just what a source says, but what supports it, what attacks it, what remains unresolved, and what it is not yet licensed to conclude.

The product is a living graph of reasoned claims that AI systems can cite, query, challenge, and extend.

The Platform

Three layers sit on one data model. The formal layer is optional and incremental: ordinary posts, comments, annotations, and documents can move into structured reasoning through guided actions, without requiring users to learn the underlying theory.

A **collaboration layer** — feeds, profiles, rooms, messaging, long-form articles, document libraries, annotations, shared source collections, live workflows — lets communities post, publish, and gather evidence without touching the formal machinery underneath. The platform is not built on behavioral tracking or engagement ranking; the valuable data is explicit reasoning activity: claims, evidence, objections, revisions, and provenance.

A **reasoning layer** turns any of that activity — a comment, an annotation, a passage, a source note — into a structured object: a claim, an argument, a scheme instance, a critical question, a challenge, a piece of evidence, a dialogue move. These objects are addressable, versioned, citable, and durable.

An **AI layer** exposes the graph to models. A single permalink resolves as whatever the caller needs: a human-readable page, structured linked data, an Argument Interchange Format graph, an attestation envelope, an embed, a social card, or an MCP artifact. The same endpoints support argument search, claim stances, counterargument discovery, citation resolution, structured authoring, critical-question answering, challenge filing, confidence folding, and deliberation-level readouts.

Those structured objects can also be embedded back into living documents, briefs, peer reviews, and knowledge-base pages, so the reasoning graph is not separate from the outputs people actually read.

The payoff is that an AI system never has to reconstruct the argumentative context around a claim. The context is already computed, stored, and citable as data.

Concretely: an agent recently posted a policy argument to Isonomia — that the EU should fund peatland rewetting through a results-based payment indexed to water-table depth. It did not land as a paragraph

or a single line of reasoning but as a graph: eleven arguments, nine connections, three levels deep, in which one keystone premise — water-table depth controls peat oxidation — branches to feed both an emissions-magnitude strand and a cost-effectiveness strand, and five strands of evidence converge on the proposal. Each premise carries its argumentation scheme and citations anchored to specific figures and pages. Two objections are wired against different targets: a land-use-displacement objection that *undercuts* the inference from on-site savings to net climate benefit, and a methane objection that *undermines* a premise of the cost-effectiveness strand. Two further claims — a counterfactual emissions estimate and a hypothetical adoption outcome — are isolated in scopes so they cannot leak into what the argument asserts. The whole object currently carries a standing of *untested, provisional until challenged*; that standing changes through the platform's challenge and evaluation workflow, not merely because an author attaches a counter-node. An agent that cites this argument inherits all of it — schemes, evidence, attack topology, scope boundaries, and current standing — rather than a URL. (Live at: <https://isonomia.app/chains/cmpvvmxqk002s8c99uxmsho4v>)

The AI Problem

The failure has three distinct shapes, and prose-generation fixes none of them.

The first is **lost topology**. A model can render a debate fluently while quietly flattening it — treating a load-bearing premise and a decorative one as equals, missing that an objection attacks the inference rather than the conclusion, ignoring an open critical question that blocks the claim outright.

The second is **coarse citation**. Models cite documents because documents are what the web offers. But the epistemic unit is smaller than a page: a single premise, a single inference, a single piece of evidence. Citing the page is like citing a library when you mean a sentence.

The third is **unstable judgment**. Asking one model to grade another's summary produces a verdict that shifts with phrasing and presentation. Isonomia's answer is to stop relying on generated judgment for the things that can be computed. Standings, refusal surfaces, critical-question coverage, provenance, and strongest counters are deterministic functions of the graph, not another layer of prose about it.

Better models do not make the external graph obsolete. They make it more valuable. Isonomia does not bet that frontier models will remain bad at structure-tracking forever; it assumes that high-stakes reasoning still needs state outside the model: transparent, inspectable, versioned, challengeable, and computable. A stronger model can help populate and revise the graph more cheaply, but it still benefits from an external record of what was asserted, who or what authored it, which evidence supports it, which objections remain open, and which conclusions the current state does not license. The durable layer is not a workaround for weak models; it is epistemic infrastructure for stronger ones.

The platform does not remove judgment from reasoning; it breaks judgment into inspectable parts. Instead of asking for a global judgment in prose, it decomposes reasoning into typed operations: extract a premise, propose a scheme, attach evidence, classify an objection, answer a critical question, surface a warrant. Models perform better under explicit structure, and their failures become local. A weak premise, mistaken attack type, or bad evidence link can be challenged and repaired without discarding the whole argument. Confidence and standing live at the joints, so the graph exposes weakest links rather than hiding them inside a fluent summary.

How the Graph Gets Populated

Isonomia does not rely on mass manual annotation after the fact. That is where earlier structured-argumentation systems broke: the formalism worked, but the capture cost was too high. Isonomia captures structure inside the workflows where reasoning is already being produced — source annotation, article drafting, peer review, policy briefing, argument search, critical-question answering, challenge filing, and AI-assisted authoring.

Models propose first-pass structure; users confirm, edit, or reject it where commitment matters; and the graph records authorship, provenance, and standing so unrati ed AI structure remains visibly provisional rather than silently becoming knowledge. The scaling bet is not that LLMs are trusted to reason for us. It is that LLMs lower the cost of structuring, while the platform determines what has standing.

The business case does not depend on a web-scale public corpus on day one. It starts in domains where structured reasoning is already worth the capture cost: research groups, policy shops, legal and regulatory analysis, scientific review, enterprise risk and compliance, grantmaking, institutional decision workflows, and AI safety and evaluation teams. The public corpus is the compounding upside, not the first proof point.

Category Positioning

Isonomia is an epistemic infrastructure layer for AI, sitting between raw content repositories and the applications that consume them. It borrows from social networks, debate sites, fact-checkers, and document editors, but it is none of them: the graph and protocol engine are the source of truth, and everything else is a surface over that.

The nearest honest descriptions are a citation and provenance layer for agents, a structured argument graph and search engine, a protocol surface for model-context clients, a deliberation-state API for high-stakes analysis, and a graph-derived guardrail for AI writing. The short version: **Isonomia is the epistemic substrate for AI agents.**

Core AI Capabilities

Machine-citable arguments. Every claim and argument resolves to a stable permalink that is not a page but a structured artifact: conclusion, premises, scheme, evidence, challenge history, current standing, authorship provenance, and citation envelope. A client cites the unit and inherits all of it — what the argument asserts, how it is supported, what evidence was fetched, what hash verifies the source, and what objection is currently strongest. This is the right granularity for research, policy, science, legal analysis, and institutional decision support, where a page-sized citation is too blunt to be accountable.

Dialectical honesty by construction. Opposition is attached to citation. A search result or argument citation returns the strongest known counterargument — or an honest null when none is on file. Standing is a named dialectical state, not an opaque score: an argument is untested, supported, attacked, undermined, survived, or blocked by unanswered obligations. Most AI citation workflows are one-sided by default; they retrieve something that supports the answer and stop. Making counters first-class makes that harder to do by accident.

Provenance and attestation. The platform hashes canonical arguments and records source-level provenance — fetched content, timestamps, content hashes, archive snapshots where available, source-confidence tiers, and unattested-premise counts. Evidence enters through a resolver that can identify DOI and scholarly metadata, enrich sources through academic indexes, and preserve reachable sources

through archive snapshots, so the graph is not just user-entered links. As AI outputs increasingly need to prove what they cited and when, a citation envelope carrying source hashes and archive links is a far stronger object than a bare URL.

A bidirectional MCP surface. Agents do more than read. Over Model Context Protocol they can orient, look up arguments, find counters, retrieve stances, resolve citations, browse schemes, propose arguments and chains and warrants, answer critical questions, and challenge answered ones. AI-authored and AI-assisted contributions are flagged at the row level, and logical standing is gated on provenance and human ratification rather than silently treating model output as equivalent to human commitment. This is not a shadow system built for LLMs — the model is a client of the same graph and API that power the product. As MCP becomes the integration layer for agentic software, Isonomia offers a structured reasoning backend rather than a bag of documents.

Deliberation-level readouts and refusal surfaces. Above individual arguments, the platform computes fingerprints of a whole deliberation: contested frontiers, missing-move reports, chain exposure, weakest-link projections, and synthetic readouts. The distinctive output is a **refusal surface** — a contract, derived from the graph, telling an AI writer what it may not assert, what it must hedge, and which honesty lines it must include, each keyed to a real graph element. This turns "be careful" into something enforceable.

Argument-native search. Isonomia searches for positions, not pages. A result is a ranked set of structured arguments carrying standing, scheme, dialectical fitness, retrieval audit data, attestation links, and discovered counters; a claim-stance query returns the for and against arguments in a single shape. Keyword search finds pages and vector search finds similar text — neither finds a structured position together with its opposition.

A formal reasoning substrate. Underneath sit several formal systems: argumentation schemes and critical questions, grounded argument evaluation, typed dialogue moves and commitment stores, Ludics designs and orthogonality, an evidence algebra, log-odds confidence folding, culprit-set belief revision, and cross-context transport. Users never have to learn any of this. The value is that AI clients can consume the outputs — grounded status, attack topology, open obligations, confidence bands, minimal disagreement loci, evidence provenance — and lean on a symbolic substrate when the task demands accountable reasoning rather than fluent guessing.

Implemented and Near-Term Capabilities

The following are implemented or substantially implemented:

- content-negotiated permalinks for structured arguments, resolving to data, graph, attestation, social-card, embed, and oEmbed surfaces;
- MCP read and write access over the argument graph;
- structured authoring by agents, with AI authorship provenance preserved;
- critical-question answering and challenge workflows;
- public argument search and claim stances;
- strongest-counter enrichment on citations;
- citation resolution with source-confidence tiers and archive fallback;
- deliberation fingerprints and contested-frontier views;
- missing-move reports and chain-exposure projections;
- synthetic readouts with honesty lines and refusal surfaces;
- graph-derived writing constraints;

- log-odds confidence algebra with one-hop cross-context evidence transport;
- Ludics substrate elements: witnessing records, articulation lattices, fossil records, event-stream primitives;
- evaluation harnesses for briefing fidelity, critical-question prioritization, and premise extraction.

These have been exercised through live MCP flows in which agents propose arguments, build branching chains, answer critical questions with session-scoped self-canonicalization, and verify idempotency and error handling.

Why This Is Not "LLM for Debate"

The crowded space is AI summarization, search, note-taking, and debate assistance. Isonomia's position is that it builds the objects those products eventually wish they had.

An ordinary research assistant ends at *"here is a summary of the debate."* Isonomia keeps going: here are the claims and their premises; here is the scheme licensing the inference; here are the critical questions still open; here is the strongest objection and whether it rebuts, undermines, or undercuts; here are the sources with their fetch and archive provenance; here is the current standing; here is the weakest link in the chain; here is what a writer must not yet assert; here is the exact state behind the conclusion.

That is the moat. Isonomia does not generate better prose over the same web — it changes the unit of retrieval, citation, and grounding.

Product Wedges

AI citation primitive. Packaged as a citation API, Isonomia lets any external agent cite structured arguments with provenance, standing, and opposition — immutable content-hash URLs, strongest-objection inclusion, source provenance and archive metadata, and model-readable standing and refusal fields. Buyers: AI research tools, enterprise knowledge assistants, legal and policy platforms, academic AI products, model-evaluation systems, and safety teams.

Argument search API. Search that returns positions rather than documents — ranked structured arguments with counters, evidence counts, scheme, standing, and stance, available over dense and sparse retrieval with quality and tested-argument filters. Buyers: research organizations, think tanks, policy and intelligence analysts, legal researchers, scientific reviewers, and any system that needs retrieval with disagreement attached.

AI briefing and refusal layer. Turn a deliberation, a document collection, or an enterprise knowledge base into a graph-derived briefing with explicit constraints: load-bearing premises surfaced, open critical questions flagged, a refusal surface naming what must not be asserted and what should be hedged, and fidelity scorecards that catch hallucinated structure. Buyers: enterprise teams needing memos with provenance, institutional decision-makers, scientific platforms, and governance, risk, and compliance groups.

Agent authoring and stress testing. Let agents propose arguments, chains, warrants, citations, and critical-question answers under strict authorship provenance and human ratification — prose-to-structure extraction, scheme suggestion, assumption surfacing, evidence-gap detection, draft quarantine, and idempotent write paths. Buyers: advanced research teams, AI writing tools, enterprise analysts, and model-assisted policy teams.

Evaluation harness for AI reasoning. The graph's manifests become a testbed for systems claiming reasoning or research competence — fixed deliberation corpora, structural ground-truth manifests, hub-set and load-bearing-premise metrics, hallucinated-structure detection, adversarial topology and prompt-injection cases, and regression testing across model and prompt changes. Buyers: AI labs, eval companies, research-tool vendors, and institutions validating AI-assisted analysis.

Future Directions

A web of machine-citable arguments. A crawlable public corpus where every claim carries support, opposition, provenance, and standing — with canonical argument identifiers, DOI-style references, cited-by graphs, browser extensions that expand Isonomia links inline, and bridges to established argument and structured-data formats.

An agent reasoning backend. The service an agent calls when it needs to know whether a claim has survived challenge, where the disagreement lives, or what it must retract to reject a conclusion — through hosted orientation contracts, citation preflight, graph-derived constraints injected into prompts, and automatic refusal when the graph blocks a conclusion.

Provenance-aware AI writing. A writing layer for reports, briefs, and reviews, with living documents that bind generated statements to live arguments, real-time attack registers on written conclusions, confidence cards explaining why a statement is licensed, snapshot-and-diff over cited states, and automated hedging keyed to standing.

Structured debate data for training and evaluation. The corpus itself — arguments with premises, schemes, evidence, critical questions, attack types, standings, provenance, and resolution trajectories — as supervised data for argument mining and scheme classification, preference data over critical-question answers, and benchmarks for counterargument retrieval and citation-grounded reasoning.

Minimal-disagreement debuggers. The Ludics layer enables a debugger for disagreement: rather than declaring two parties globally opposed, it locates the minimal unshared commitment, or the minimal set of divergence points in branching cases — powering critique constrained to the actual divergence locus and agent tools that ask the single shortest question needed to resolve a dispute.

Cross-context knowledge transport. The Plexus network moves arguments between rooms, organizations, and knowledge spaces with provenance, confidence bands, and one-hop auditability — enabling institutional knowledge reuse, cross-context claim mapping, transport confidence gates, and privacy-preserving federated search.

Evidence algebra as a trust substrate. Log-odds folding and culprit-set revision give a lawful alternative to opaque confidence scores: signed pro/con aggregation, queries asking what must be retracted to reject a claim, local, imported, and total confidence bands, monotonicity tests, and model-readable confidence explanations.

Defensible Moats

The data model. The core object is a structured argument graph — claims, premises, schemes, critical questions, attacks, evidence, commitments, confidence, provenance, live state — not a document, thread, or vector. That is very hard to bolt onto an ordinary content product after the fact.

The protocol. The model surface is bidirectional. Agents read, propose, answer, challenge, and cite while the system holds idempotency, authorship provenance, and graph integrity. It is not a one-way export.

The formalism. Structured argumentation, critical questions, typed dialogue moves, commitment stores, interchange formats, evidence algebra, confidence folding, and Ludics semantics combine into machine outputs that are richer and more enforceable than anything a prose pipeline produces — and almost all of it stays invisible to users.

The evaluation. Fidelity harnesses are treated as product infrastructure: ground-truth manifests are computed from the graph and used to grade model outputs mechanically, which beats manual spot-checks over sample summaries.

The provenance. The platform records not only what is claimed but how it entered, who or what authored it, which tool created it, what it cites, what state it was retrieved in, and what objections existed at the moment of citation.

The corpus. Deployed at scale, Isonomia accumulates something no one else has: reasoning *trajectories* — attacks, revisions, warrants, concessions, retractions, and evidence updates tracked over time.

The Formal Moat

This section is for readers conducting technical diligence — AI labs, evaluation vendors, and institutions validating AI-assisted reasoning. It presumes familiarity with structured argumentation and Ludics; general readers can skip it without losing the argument above. Proofs referenced here are available for review under a cross-check policy.

The formalisms Isonomia builds on are public. ASPIC+, the Walton scheme taxonomy, the Argument Interchange Format, Girard's Ludics, log-odds confidence folding, and Ambler's categorical evidence semantics are all in the literature.

The defensible layer is the original work above the inheritance. The results below are necessary for a number of the product surfaces in the platform. Each entry states what the theorem buys and why a reimplementing that lacks it ships a subtly wrong product.

Certification is a theorem, not an analogy (T005). Grounded acceptance in a finite argumentation framework corresponds, through a strategy-preserving bijection, to acceptance-by-interaction in multiplicative, additive-free Ludics — Proponent strategies to Proponent designs, orthogonality to convergence. This is what licenses "orthogonality testing is the certification procedure" as a proven equivalence to grounded-extension status rather than a heuristic that resembles one. A competitor can imitate the output; matching the *guarantee* requires the bridge. For a buyer asking whether a "survived" badge means anything procedurally, this is the claim under it.

Minimal disagreement on branching disputes (T006–T009). Over concession trees, the minimal separating context — the smallest set of points where two positions first diverge — is the antichain of per-line first-divergence loci under the Smyth (upper-powerdomain) order; under the Hoare order no minimum exists at all. The consequence is sharp: a reimplementing using a plausible-but-wrong notion of "minimal disagreement" returns incorrect loci on branching disputes, which are the common case, not the corner case. The minimal-disagreement extractor is correct because the order-relative result is proven; a heuristic version is silently wrong exactly where it is most often used.

The refusal surface rests on a geometric fact (T001–T002). A behaviour decomposes into disjoint cones, and its incarnation set is an antichain, which yields a hard structural constraint — aggregation cannot cross cones. The refusal surface that tells an AI writer what it may not assert is computed from that constraint. Without the decomposition the constraint is a heuristic that can be violated without anyone noticing, which is the failure mode a refusal surface exists to prevent. (The original *global* join-semilattice version of this result is false; only the per-cone version holds — see below.)

Multi-hop transport safety (T010). Room-to-room transport functors form a bicategory; the transport map is a pseudofunctor exactly on the monodromy-free sub-bicategory, and multi-hop transport is sound iff no claims drop or drift across the path. This converts "one-hop, auditable to a single source room" from a contract assertion into a theorem about precisely when more than one hop is safe.

The map of dead ends is itself the asset. A copyist working from the public description inherits none of the negative knowledge this program paid for, and two cases are load-bearing. The original global join-semilattice claim is false; only the per-cone result (T001) holds, so a reimplementation reading the surface description re-derives the false version. Minimality across opponents was refuted in its first form and recovered only under a proper-test redefinition (T008–T009). The record of what does not work, and why, exists only here and is expensive to reproduce — and unlike the theorems, it confers no benefit to anyone who has not already done the work.

What is not claimed. The program has an active research frontier — among others, a candidate fourth, *structural* attack type for the case where two schemes have disjoint behaviours over the same claim and no rebut/undermine/undercut applies, and the confidence-graded, cartesian-closed remainder of the Ludics $\rightleftharpoons$ Ambler bridge. These are owned open questions, stated with their settlement conditions, not results. Nothing in the moat above rests on them, and they are listed here to mark the boundary between what is proven and what is conjectured.

How to check this without reading the proofs. Pre-registered ablation studies convert formal correctness into a measurable output difference, which is the form in which an evaluation team can verify it directly: a pattern-versus-content ablation on scheme classification, inter-annotator agreement conditioned on identification-pattern match against a published κ baseline, and cone-coverage as a convergence predictor — a quantity that only exists because of the per-cone decomposition. These are the instruments by which a buyer confirms that the theorems produce better outputs rather than taking the theorems on faith.

Why Now

Three shifts make the timing right. Agents are moving from answering questions to acting across tools, and they need structured knowledge APIs rather than document search. The citation problem is becoming acute as enterprises, researchers, and regulators demand outputs that show provenance, opposition, uncertainty, and current standing. Retrieval will keep improving, but better retrieval alone still returns content, not accountable reasoning state: it can find similar passages, but it does not by itself say whether a claim survived challenge, whether a source supports a premise, or whether an inference is undercut. The next layer is retrieval plus dialectical state — which is what Isonomia is.

Closing Position

AI needs epistemic infrastructure, not just more context. The hard parts that AI companies tend to discover they need only later — structured claims and arguments, challenge topology, evidence

provenance, content-hashed citations, agent-readable APIs, graph-derived constraints, and fidelity harnesses graded against mechanical ground truth — already exist here. The collaboration and publishing layers matter because they generate real human reasoning data, but the infrastructure is the center of the opportunity: the reasoning layer beneath AI research, writing, search, enterprise knowledge, policy, science, legal argument, and institutional decision support.

Isonomia makes arguments citable as objects rather than paragraphs, attaches provenance, standing, and opposition to every citation, and tells a model what the current state of reasoning does and does not license. It turns contested knowledge into an API.